

Metabolism

1- A panel of the following tests (LDH1, LDH2, AST, CKMB) are more useful for diagnosis of.....

Gallbladder stone

Hepatitis



Myocardial infarction

Liver Cirrhosis

2- Which of the following ketonbodies is exhaled?

Acetoacetate



Acetone

Lactate

Hydroxybutyrate

3- Severe combined immunodeficiency (SCID) is caused by which of the following enzyme deficiencies?



Adenosine deaminase

Guanase

Guanase

Purine nucleoside phosphorylase

4- Histidine is degraded to α -ketoglutarate and is described as a

Keto-gluco amino acid

Gluco amino acid

Ketogenic amino acid



Glucogenic amino acid

5- Which of the following are correct about bile acids productions ?

Bile acids are produced in liver and adipose tissue

21- α -hydroxylase is the rate-limiting step



Increased secretion of cholesterol into bile causes gallbladder stones

Expression of the rate-limiting enzyme is upregulated by bile acids

6- All of the following could be the consequences of the carnitine consumption as a supplement except:

- Decreases in fatty acid synthesis in the liver
- ✓ Accumulation of fat droplets in adipose tissue
- Increases in ketone bodies production
- Increase in blood glucose concentration

7- Which tissue is the main source of Glycerol for gluconeogenesis?

- ✓ Adipose
- Heart muscle
- Liver
- Skeletal muscle

8- The synthesis of glucose from glycerol is called :

- Glycolysis
- Krebs cycle
- ✓ Gluconeogenesis
- Saponification

9- Which of the following would NOT be expected to contribute to hyperuricemia (gout)?

- ✓ Inhibition of xanthine oxidase
- High activity of adenosine deaminase
- Unusually high turnover of nucleic acids
- Unusually high levels of PRPP

10- All of the following sentences are correct about cholesterol synthesis except

- Cholesterol synthesis is decreased when ATP availability is decreased
- ✓ The statin drugs are non-competitive inhibitors of HMG CoA reductase
- The statin drugs are structural analogs of HMG CoA
- An increase in epinephrine inhibits synthesis of cholesterol

11- Net generation of energy on complete oxidation of a 13-carbon fatty acid is...

- ✓ Less than the energy generation from a 12-carbon fatty acid

Equal to the energy generation from a 14-carbon fatty acid

In between the energy generation from a 12-carbon fatty acid and an 14-carbon fatty acid

Equal to the energy generation from a 12-carbon fatty acid

12- Which of the following does not participate in the de novo synthesis of purine ring of IMP?

THF-one carbon unit

Glycine

Glutamine

✓ Ammonia

13- β -Aminoisobutyrate is the end product of degradation of which of the following?

✓ Thymine

cytosine

Guanine

Uracil

14- Transamination is the transfer of an ...

Amino acid to a keto acid plus ammonia

✓ Amino group from an amino acid to a ketoacid

Amino group from an amino acid to a carboxylic acid

Amino acid to carboxylic acid plus ammonia

15- If all potential sources of ATP production are taken into account, the net number of ATP molecules formed per molecule of glucose in anaerobic glycolysis is

36

✓ 2

6

18

16- A 7-year-old boy suffers from mental retardation and self-mutilation and has increased levels of serum uric acid. The symptoms are characteristic of Lesch Nyhan syndrome. Which of the following is defective?

De novo synthesis of pyrimidines

Xanthine oxidase

- ✓ HGPRT
Salvage pathway for pyrimidine biosynthesis

17- The end product of glycolysis is -----?

- Citric acid
- Glycogen
- Glucose
- ✓ Pyruvic Acid

18- Which glucose transporter is active in muscle tissues?

- GLUT-1
- GLUT-3
- ✓ GLUT-4
- GLUT-2

19- The major control of de novo pyrimidine nucleotide synthesis in man is:

- Availability of N-acetyl glutamate
- Substrate availability
- ✓ Competitive inhibition of carbamoyl phosphate synthetase II
- Feedback inhibition of glutamine-PRPP amidotransferase

20- The atherogenic Lipoprotein is a collection of which lipoprotein abnormalities?

- Reduced levels of HDL, high proportion of LDL and normal triglycerides
- Reduced levels of HDL, low proportion of LDL and raised triglycerides
- ✓ Reduced levels of HDL, high proportion of LDL, and raised triglycerides
- Raised levels of HDL, high proportion of LDL and raised triglycerides

21- Which of the following is not a the Kreb's cycle intermediate?

- Succinate
- Citrate
- ✓ Glucoronate
- Oxaloacetate

22- Which of the following statements about the long-term fasting is correct?



Muscles uses ketone bodies from fatty acids

The adipose tissue can use ketone bodies from its fat stores

The brain and RBCs release ketone bodies

The liver uses ketone bodies to produce energy

23- Which of the following rules regarding the interconversion of fuels are central to the integration of metabolic pathways?

Amino acids can be made from and converted to fatty acids



Fatty acids can be made from but not converted to carbohydrates and amino acids

Fatty acids can be made from and converted to carbohydrates and amino acids

Carbohydrates can be made from and converted to amino acids and fatty acids

24- Following the consumption of a mixed meal, all of the following is correct, except...

Consumption of a meal leads to suppression of lipase activity within adipose tissue



Adipose tissue lipoprotein lipase is activated by insulin

During the postprandial period, the VLDL synthesis pathway is suppressed in favour of hydrolysis of chylomicrons

Adipose tissue lipoprotein lipase is inactivated by insulin

25- A positive nitrogen balance occurs:

In advanced cancer

Following (during) surgery

In kwashiorkor



In growing infant

26- Which of the following inhibits acetyl CoA carboxylase, a rate-limiting enzyme of fatty metabolism?

Citrate

ATP



Malonyl CoA

Acyl CoA

27-

Which of the following enzymes is responsible for the conversion of pyruvate to phosphoenolpyruvate (PEP)?

- ✓ Carboxykinase
- Hexokinase
- Pyruvate dehydrogenase
- Enolase

28- Which of the following compounds is a member of the electron transport chain?

- Palmitoyl carnitine
- ✓ Coenzyme Q
- NADH
- Carnitine

29- Oxidative deamination of an amino acid is the conversion of an ...

- Amino group from an amino acid to carboxylic acid
- ✓ Amino acid to a keto acid plus ammonia
- Amino group from an amino acid to a keto acid
- Amino acid to a carboxylic acid plus ammonia

30- High level of uric acid in blood is also called?

- BUN
- Uremia
- Azotemia
- ✓ Uricemia

31- During the conversion of glucose to pyruvate, two NADH molecules are generated. Which of the following steps generates NADH? Conversion of

- ✓ glyceraldehydes-3-phosphate to 1-3-bisphosphoglycerate
- fructose-6-phosphate to fructose-1-6-bisphosphate
- phosphoenolpyruvate to pyruvate
- 3-phosphoglycerate to 2-phosphoglycerate

32- Thioredoxin is involved in the

- ✓ Conversion of a ribonucleotide to a deoxyribonucleotide
- conversion of dUMP to dTMP
- degradation of nucleoprotein
- conversion of AMP to ATP

33- How many ATP molecules are produced when a molecule of acetyl CoA is oxidized through citric acid cycle?

- 38
- 15
- ✓ 12
- 24

34- Where does TCA cycle take place?

- Cell membrane
- Cytosol
- ✓ Mitochondria
- Nucleus

35- A person with phenylketonuria cannot convert ...

- Phenols into ketones
- ✓ Phenylalanine to Tyrosine
- Phenylalanine to lysine
- Phenylalanine to isoleucine

36- All of the following are correct about liver transaminases except....

- ✓ AST elevations persist longer than do ALT
- ALT is more liver specific than AST
- Major portion of AST in serum is of mitochondrial origin
- Patients with cirrhosis often have normal or only slightly elevated serum AST and ALT levels

37- Creatinine is the product of..... creatine or phosphocreatine in muscles.

- Hydration-Dehydration
- Reduction-Phosphorylation

Oxidation-Reduction



Dehydration-Dephosphorylation

38- All of the following are correct about introducing double bonds into fatty-acyl chains except

Desaturase is responsible for the reaction

Smooth Endothelial Reticulum is responsible for desaturating LCFA



Δ^{12} double bond is inserted

Cis double bond is inserted

39- The key regulatory enzyme of cholesterol synthesis is.....

HMG Co-A Synthase



HMG Co-A reductase

HMG Co-A Lyase

7- α Hydroxylase

40- Most of the ammonia released from L- α -amino acids reflects the coupled action of transaminase and

Serine dehydratase

Histidase

L-amino acid oxidase



Glutamate dehydrogenase

41- All of the following is correct about fat digestion, except

Secretin causes the pancreas to release a solution rich in bicarbonate

Orlistat inhibits both gastric and pancreatic lipases



Cholecystokinin (CCK) increases gastric motility

Cholecystokinin (CCK) is released in response to the presence of

lipids in the upper small intestine

42- The first substance produced in the citric acid cycle is:



Citrate

Acetyl CoA

ATP

Oxaloacetate

43- Which group of amino acids is required for creatine formation?

Serine, glycine, methionine

Glycine, methionine, leucine

Phenylalanine, arginine, tyrosine

✓ Glycine, arginine, methionine

44- The connecting link between Hexose Mono Phosphate shunt and lipid synthesis is -----.

✓ NADPH

NADH

Ribose

Sedoheptulose 7 -phosphate

45- All of the following enzymes in glycolysis catalyze an irreversible reaction except

Phosphofructokinase

Hexokinase

✓ Phosphoglycerate mutase

Pyruvate kinase

46- Inhibiting the rate limiting enzyme of cholesterol biosynthesis would be beneficiary in lowering blood cholesterol levels by one of the following mechanisms:

✓ Inhibiting the formation of HMG-CoA

Preventing the reabsorption of bile acids from the intestine

Inhibiting the absorption of cholesterol from the diet

Increasing the conversion of cholesterol to bile acids

47- Which of the following statements is incorrect regarding fatty acid β -oxidation?

Peroxisomes are the site for the degradation of xenobiotics and eicosanoids

✓ The major site of fatty acid β -oxidation is the peroxisomes, the mitochondria also contain enzymes for this pathway

The availability of fatty acids and the rate of utilization of the β -oxidation products regulate the rate of β -oxidation

The liver peroxisomes oxidize very long chain fatty acids to medium chain products

48- The 2 nitrogen atoms in urea are contributed by

Ammonia and glutamate

Ammonia and alanine



Ammonia and aspartate

Glutamine and glutamate